

KIPRE STAFF TRAINING NEEDS ASSESSMENT (TNA) REPORT

Human Resource Division

Prepared by CStat. Patrick Mwaura (DSAS)

2026-06-08

Contents

Executive Summary	2
List of Acronyms and Abbreviations	3
Background	4
Objectives	4
Methodology	5
Results	5
Workforce profile	5
Objective 1: Perceived skill gaps	7
Objective 2: Perceived impact of past training	11
Objective 3: Perceived Training priorities	12
Objective 4: Training delivery preferences	12
Objective 5: Perceived workplace bottlenecks	12
Objective 6 : Innovation and innovation readiness	12
Conclusion and Recommendations	12
Key plots	12

Executive Summary

Table 1: Executive Summary Dashboard: Synthesis of Core Training Impact and Competency Skews

Focus Area	Key Insight & Strategic Action Mandate
Workforce profile	87.5% of voluntary survey respondents cluster tightly within five core operational grades (NM7–NM12), heavily driven by NM8 as the single largest cadre at 28.7%. Structural promotion bottlenecks are a highly visible concern among active respondents.
Workforce experience	An extensive 36.2% of the active workforce reports a tenure of 16+ years. This reflects a profound self-selection survey bias where veteran staff felt a greater personal stake in completing the optional questionnaire than incoming cohorts.
Training effectiveness	Initial induction courses yield peak value for new entries (4.80/5.00 effectiveness score). However, a critical systemic drop occurs in the 1–3 years bracket, where scores plummet to a mediocre 2.92, marking the only group where skill absorption is non-unanimous (92.3%).
Competency profile	High baseline interpersonal competencies like Communication and Customer Service (both at 55.0% High) mask a critical organizational vulnerability: a massive 59.7% KIPRE-wide majority of competencies sit trapped on the Medium Proficiency warning track.
ICT capability gap	Digital infrastructure skills represent the organization’s primary bottleneck. Within the intermediate 1–3 years cohort, a strong 56.2% claim high technical capability, yet a meager 18.8% possess advanced ICT skills, confirming that intermediate training ignores modern workflows.
Long-tenure workforce	A severe digital divide separates senior cadres from emerging staff. Veteran personnel with 16+ years of service exhibit an extreme technological deficit, with a mere 10.3% retaining high digital capabilities, driving deficiency severity to an institutional peak of 1.02 within grade NM11.
Directorate variation	Capability gaps are highly localized rather than uniform. The ASWED directorate operates as the primary departmental risk center, generating the worst cumulative deficiency profile in the organization, led by an acute severity index of 1.07 in ICT.
Division variation	Granular mapping isolates the WED division as a severe structural bottleneck, exhibiting highly deficient clusters across Information Systems (1.02), Leadership (0.89), and Policy (0.89), followed closely by intermediate leadership strains in ECEHD (0.94).
Training priority	Pivot away from generic corporate workshops. Capacity-building pipelines must be aggressively directed toward advanced digital systems literacy, M&E analytical frameworks, and regulatory Policy Implementation, specifically targeted at the NM8/NM11 cadres, ASWED directorate, and WED division.

List of Acronyms and Abbreviations

Table 2: Institutional Acronyms and Organizational Definitions Mapping

Acronym	Classification	Full Institutional Definition
ASWED	<i>Directorate</i>	Animal Sciences Welfare and Ethics Directorate
CBPGMD	<i>Directorate</i>	Capacity Building Partnership and Grant Management Directorate
CCD	<i>Division / Section</i>	Corporate Communication Division
CSD	<i>Directorate</i>	Corporate Services Directorate
CSLSD	<i>Directorate</i>	Corporation Secretary Legal Services Directorate
DSAS	<i>Division / Section</i>	Data Science and Analytics Section
EECEHD	<i>Division / Section</i>	Evolutionary Ecology Conservation and Ecosystem Health Division
FAD	<i>Division / Section</i>	Finance and Accounts Division
HRAD	<i>Division / Section</i>	Human Resource and Administration Division
IARAD	<i>Directorate</i>	Internal Audit and Risk Assurance Directorate
ICTD	<i>Division / Section</i>	Information Communication Technology Division
IDOHD	<i>Division / Section</i>	Infectious Diseases and One Health Division
KSRID	<i>Division / Section</i>	Kenya Snakebite Research and Intervention Division
LSS	<i>Division / Section</i>	Laboratory Services Section
NCDD	<i>Division / Section</i>	Non Communicable Diseases Division
NLABD	<i>Division / Section</i>	National Laboratory Animal Breeding Division
ODG	<i>Directorate</i>	Office of the Director General
PGMD	<i>Division / Section</i>	Partnerships and Grants Management Division
PSD	<i>Division / Section</i>	Planning and Strategy Division
RHBD	<i>Division / Section</i>	Reproductive Health and Biology Division
RPDD	<i>Directorate</i>	Research and Product Development Directorate
VCDS	<i>Division / Section</i>	Veterinary Clinical and Diagnostics Services Division
WED	<i>Division / Section</i>	Welfare and Ethics Division

Background

The primary purpose of this Training Needs Analysis (TNA) is to systematically assess institutional competency gaps and evaluate workforce preparedness. This evidence-based approach is designed to inform training priorities and guide strategic human capital development across the KIPRE. The assessment was conducted using a multi-dimensional questionnaire structured into specific modules to capture a holistic view of the workforce. These sections included:

- Basic Information (Directorate, Age, and Sex of the employee)
- Job Role and Responsibilities
- Self Performances Assessing Questions
- Skills and Copetency Assessment
- Training History
- Identification of Training Needs
- Training Delivery Preferences
- Organizational Support
- Future Skills and Emerging Needs
- Additional Comments

By triangulating data from these domains, the analysis identifies not only current skill deficiencies but also the organizational support structures and delivery formats required to bridge them effectively.

Objectives

The primary goal of this assessment was to capture KIPRE employees' perspectives on their work environment, competencies, and developmental needs. By analyzing these self-reported insights, KIPRE can move away from ad-hoc training decisions and build a people-first human capital strategy. Specifically, this assessment aimed to:

- Map perceived skill gaps by uncovering where staff across different directorates, divisions, and job grades feel they lack the technical or administrative confidence to excel.
- Evaluate the felt impact of past training by documenting how effective staff believe previous workshops were and capturing the personal or structural hurdles they feel prevent them from using those skills.
- Capture bottom-up training priorities by listening directly to frontline operational needs and career progression goals to build a highly responsive institutional training roadmap.
- Tailor training delivery to staff preferences by identifying the exact learning formats, durations, and schedules that employees feel would best fit their workflows.
- Identify employee-perceived workplace bottlenecks by revealing the cultural or structural friction points that staff believe are currently limiting their productivity.
- Gauge readiness for future innovations by exploring how prepared and confident the team feels regarding emerging scientific and technological trends in research.

Methodology

The assessment was digitized via ODK Central and deployed from 20th – 28th April of 2026. To ensure we captured a representative voice, we used a multi-channel approach—utilizing official email and professional WhatsApp groups. Participation was voluntary, positioned as a collaborative effort to co-design the organization’s future training roadmap. Data Processing & Analysis Once the collection window closed, raw data was ingested into R (v4.5.2). The analytical pipeline focused on cleaning, recoding qualitative responses for statistical depth, and aggregating the results to provide a granular view of needs across all Directorates and Divisions.

Results

Workforce profile

A total of 80 employees, representing 42% of the 189 staff members currently at KIPRE, participated in this assessment. This respondent base spans 23 distinct divisions (see Table 3) across seven primary Directorates, ensuring the training roadmap reflects both the core scientific mandate and the essential administrative backbone of the institute. The distribution reveals a workforce concentrated within key technical hubs, most notably the Animal Sciences Welfare and Ethics and Research and Product Development Directorates. In the research domain, the Kenya Snakebite Research and Intervention (KSRIC) Division and the Welfare and Ethics (WE) Division emerged as the most represented units. This engagement from frontline technical staff—supported by a demographic where 51.2% of respondents are aged between 25 and 44—provides a robust foundation for identifying gaps in laboratory biosafety and modern research protocols. While the gender distribution is relatively balanced (see Table 4) (45% female, 55% male), distinct generational trends are evident: the female cadre is most concentrated in the 35–44 age bracket (36.1%), whereas the male workforce peaks in the 45–54 bracket (34.1%). Administrative and strategic functions are equally well-mapped, with consistent representation from Corporate Services and the Office of the Director General. Notably, only 10% of respondents are aged 55 or older, indicating a predominantly mid-career workforce with significant long-term growth potential. This demographic profile highlights a strategic opportunity to implement cross-cutting training programmes that standardise administrative excellence and ensure the institution is prepared for future succession and evolving digital demands.

Based on the voluntary sample of 80 respondents, the data highlights a heavily concentrated middle-tier profile where a staggering 87.5% of all participants belong to just five core grades (NM7, NM8, NM10, NM11, and NM12) (see Table 5). This cluster is overwhelmingly driven by NM8, which represents the single largest respondent segment at 28.7% (23 individuals), followed closely by NM10 at 15.0%. Long-tenured staff are heavily represented, with over a third (36.2%) of the entire sample reporting 16 or more years of service. While this high veteran participation indicates deep institutional memory, its concentration within specific cohorts—such as NM11 where 80% of respondents have 16+ years of tenure—likely reflects a strong self-selection bias where long-serving employees felt a greater stake in completing the survey than newer hires. Furthermore, the active participation of both entry-level staff and 16+ year veterans within identical grades (such as NM8 and NM12) indicates that structural bottlenecks or career stagnation are highly visible and sensitive topics among the voluntary respondents.

Table 3: Workforce distribution by directorate and division

Division / Section	Staff Count
Animal Sciences Welfare And Ethics Directorate	
Kenya Snakebite Research And Intervention Division	1
Laboratory Services Section	1
National Laboratory Animal Breeding Division	4
Veterinary Clinical And Diagnostics Services Division	9
Welfare And Ethics Division	11
Capacity Building Partnership And Grant Management Directorate	
Human Resource And Administration Division	1
Partnerships And Grants Management Division	3
Corporate Services Directorate	
Corporate Communication Division	4
Finance And Accounts Division	4
Human Resource And Administration Division	5
Information Communication Technology Division	1
Planning And Strategy Division	1
Corporation Secretary Legal Services Directorate	
Infectious Diseases And One Health Division	1
Internal Audit And Risk Assurance Directorate	
Human Resource And Administration Division	1
Office Of The Director General	
Human Resource And Administration Division	1
Planning And Strategy Division	1
Research And Product Development Directorate	
Data Science And Analytics Section	1
Evolutionary Ecology Conservation And Ecosystem Health Division	7
Infectious Diseases And One Health Division	7
Kenya Snakebite Research And Intervention Division	10
Laboratory Services Section	1
Non Communicable Diseases Division	1
Reproductive Health And Biology Division	4

Table 4: Gender distribution across age brackets

Gender	Under 25 years	25-34 years	35-44 years	45-54 years	55+ years	Total
female	8.3% (3)	22.2% (8)	36.1% (13)	27.8% (10)	5.6% (2)	100.0% (36)
male	6.8% (3)	29.5% (13)	15.9% (7)	34.1% (15)	13.6% (6)	100.0% (44)
Total	7.5% (6)	26.2% (21)	25.0% (20)	31.2% (25)	10.0% (8)	100.0% (80)

Table 5: Job grade distribution by length of service (Voluntary Survey)

Job_Grade	Less Than 1 Year	1 3 Years	4 7 Years	8 15 Years	16+ Years	Total	Percentage (%)
NM1	0	1	0	0	0	1	1.2
NM2	0	1	0	0	0	1	1.2
NM5	0	0	0	1	3	4	5.0
NM6	0	0	0	0	2	2	2.5
NM7	0	3	2	2	3	10	12.5
NM8	4	3	9	3	4	23	28.7
NM9	0	0	0	1	1	2	2.5
NM10	0	1	2	4	5	12	15.0
NM11	0	0	1	1	8	10	12.5
NM12	2	5	1	1	2	11	13.8
NM13	0	2	0	1	1	4	5.0

Objective 1: Perceived skill gaps

The baseline institutional data establishes that while the organization operates from a fundamentally stable footing, it faces a pervasive mid-level capability ceiling. Across all evaluated competencies, the workforce maintains a baseline Average Severity Score of 0.662 (see Table ??), with a healthy 37.0% of responses indicating High Proficiency (Safe Zone) and only a minimal 3.3% falling into the Low Proficiency (Critical Alert Zone).

Average Severity Score	High Proficiency (%)	Medium Proficiency (%)	Low Proficiency (%)
0.66	37.00	59.70	3.30

Table 6: Overall Institutional Competency Gap Profile

However, a deeper look into specific competency areas reveals that this stability is heavily anchored by general operations (see Figure 1), such as Communication (55.0% High), Customer Service (55.0% High), and baseline Technical applications (48.8% High). Once these foundational administrative strengths are set aside, a critical structural vulnerability emerges: a massive 59.7% company-wide majority of competencies sit clustered within the Medium Proficiency (Warning Zone). This exposure is most acute within the organization’s specialized technical and regulatory pillars—specifically ICT and Policy Implementation—where 72.5% and 71.2% of the workforce, respectively, reside on this warning track.

This profile diagnoses an organization whose staff possess excellent interpersonal and day-to-day administrative skills, but encounter a clear operational ceiling when tasked with handling advanced digital infrastructure or complex regulatory frameworks. Rather than pointing to a widespread institutional crisis, the data uncovers a highly focused training mandate. To breach this capability ceiling, strategic development resources must pivot away from generic corporate workshops. Instead, interventions should be aggressively channeled into formalizing advanced capacity-building pipelines for complex ICT systems management, Monitoring & Evaluation (M&E), and specialized Policy Implementation.

Directorate level Competency Gap Profiling The shrinkage-adjusted matrix illustrated in Figure 2 expands the institutional capability diagnostics by mapping self-reported proficiency profiles across the organization’s distinct Directorates. This high-level view proves that operational vulnerabilities are highly

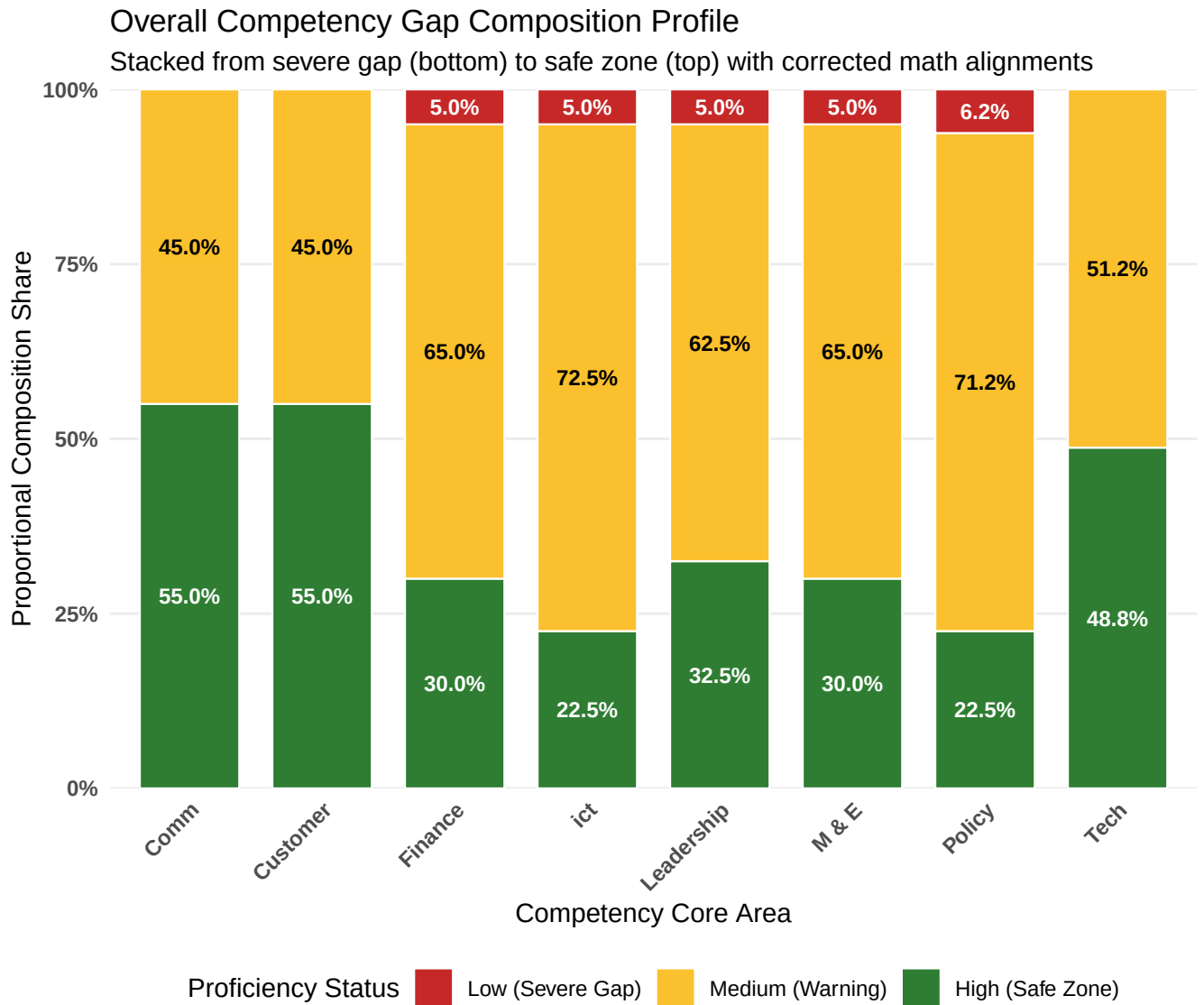


Figure 1: Overall Competency Gap Composition Profile by Core Area.

localized, driven by specific functional mandates rather than a generalized systemic failure. Because these metrics depend on self-reporting, they express the relative comfort levels and perceived training needs of teams within their specific administrative workflows.

The most prominent institutional hotspot is concentrated in the **ASWED** directorate. It presents the highest self-reported deficiency score across the entire organization, peaking at an acute **1.07 in ICT**—where a notable 19% of its respondents explicitly state a low proficiency severe gap. This directorate further reports heightened constraints in Finance (0.88), Leadership (0.88), and Policy applications (0.85). This profile strongly contrasts with more operationally secure directorates, such as **RPDD** and **CSD**, which maintain robust, low-severity baselines across core administrative indicators like Communication and Customer Service.

Concurrently, a uniform horizontal warning track spans nearly all departments in the areas of **Policy** and **M&E**, reinforcing earlier findings that regulatory compliance and analytical tracking are broad institutional anxieties. Ultimately, these self-reported baselines show that technical upskilling resources should not be distributed equally across the organization; instead, advanced ICT and financial management pipelines must be aggressively directed toward the **ASWED** directorate to resolve localized operational strain.

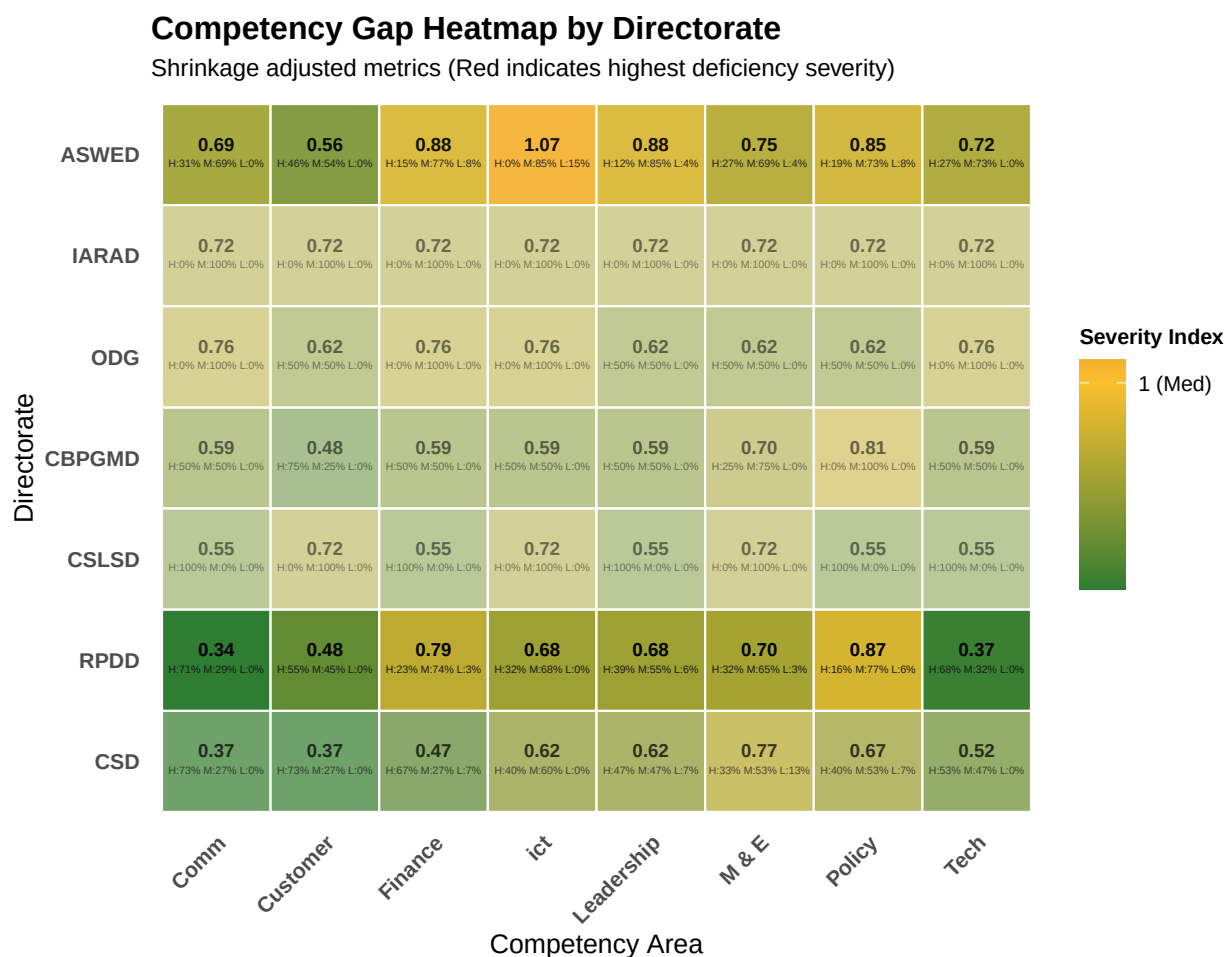


Figure 2: Competency Gap Heatmap by Directorate Cadre.

Divisional level Competency Gap Profiling The shrinkage-adjusted division matrix illustrated in Figure 3 provides the most granular view of the organization's capability profiles. By evaluating these localized self-reported scores, the data pinpoints exactly where operational vulnerabilities reside within the corporate matrix, reflecting the distinct day-to-day anxieties and perceived capacity constraints of individual working teams.

The most critical operational exposure is highly concentrated within the **WED** division. This unit displays a severe cluster of high-deficiency self-assessments, peaking at an acute **1.02 in ID** (where 18% of respondents report an absolute low proficiency severe gap), **0.89 in Leadership**, and **0.89 in Policy**. Minor, yet notable secondary warning nodes are visible within **EECEHD** (peaking at **0.94 in Leadership**) and **KSRID** (elevated at **0.89 in Policy**). Conversely, operational core units like **FAD**, **NCDD**, and **ICTD** present robust, deep green baselines, expressing high levels of comfort across nearly all administrative indicators.

Ultimately, these granular subjective findings prove that institutional deficiencies are not systemic across all working teams. Instead, they uncover a highly surgical upskilling roadmap, dictating that intermediate leadership development and policy infrastructure pipelines must be aggressively funneled into the **WED** and **EECEHD** divisions to eliminate localized capability gaps.

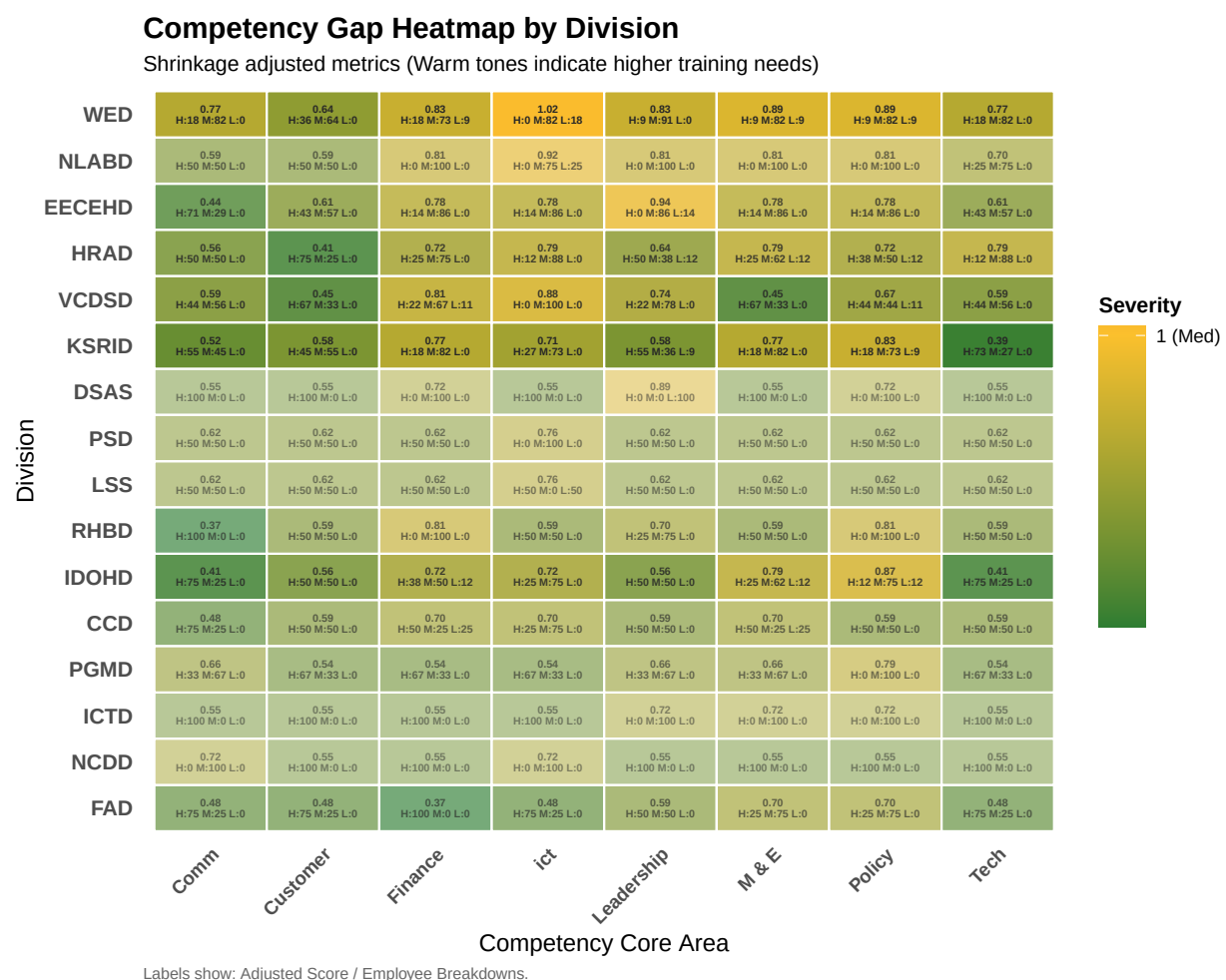


Figure 3: Competency Gap Heatmap by Division.

Cadre-Level Competency Gap Profiling The shrinkage-adjusted matrix illustrated in Figure 4 clarifies that the organization’s mid-level capability ceiling is driven by critical, localized vulnerabilities rather than a uniform institutional deficit. Because these metrics are entirely self-reported, they reflect the employees’ personal comfort levels and perceived operational limitations within their specific roles.

The most severe self-reported operational risk is heavily concentrated within the senior operational margin of grade **NM11**, which displays the highest deficiency severity scores in the organization—peaking at a critical **1.02 in ICT** (where 20% of respondents judge themselves as having a severe gap), **0.95 in Policy**, and **0.89 in Monitoring & Evaluation (M&E)**. This directly highlights an acute sense of technological and regulatory isolation among the long-tenured veteran staff who dominate this cadre.

Concurrently, a highly visible warning track runs through the organization’s primary operational layers, specifically within **NM8** and **NM12**. While **NM8** staff report high confidence in interpersonal competencies like Communication and Customer Service, they acknowledge a clear capability ceiling when handling specialized inputs, reporting elevated deficiency indices in **Policy (0.73)** and **ICT (0.62)**. Because **NM8** represents the single largest segment of the voluntary sample (28.7%), this widespread self-reported vulnerability confirms that digital and regulatory gaps directly impact core service delivery.

Ultimately, these subjective baselines indicate that capacity-building resources must pivot away from generalized workshops and instead focus on technical upskilling in digital systems literacy and policy applications targeting the **NM8, NM10, and NM11** cadres.

Objective 2: Perceived impact of past training

Table 7 presents an integrated profile of capacity-building impact and current proficiency levels across employee tenure brackets. Out of 80 voluntary respondents, 56 (70.0%) have attended past training, creating a solid baseline for analysis. The data reveals a sharp, non-linear drop in performance for early-career staff. New hires with less than a year of service report excellent results, with an effectiveness score of 4.80 out of 5.00 and a 100% skill acquisition rate. However, for the 1–3 Years cohort, average training effectiveness plummets to 2.92, making it the only group where skill acquisition is not unanimous (92.3%). The table reveals the root cause of this drop: while 56.2% of the 1–3 years cohort report high technical skills, a meager 18.8% possess high ICT skills. This major difference proves that intermediate training focuses too much on routine manual tasks and neglects modern digital workflows. A different structural issue affects long-serving staff (16+ Years). This group actively participates in training ($n = 18$) and rates it highly (3.94), yet only 10.3% have high ICT capabilities. This shows that while mid-career training successfully builds capabilities up to the seven-year mark (peaking at 73.3% technical and 40.0% ICT proficiencies), the organization lacks the continuous digital upskilling needed to prevent early-career bottlenecks and long-tenure technological isolation.

Table 7: Breakdown of Training Impact and High Skill Proficiency by Tenure

Service Tenure	Sample (N)	Attended (n)	Avg Eff. (1-5)	Skill Acq. (%)	High Tech (%)	High ICT (%)
Less Than 1 Year	6	5	4.80	100.0%	33.3%	33.3%
1 3 Years	16	13	2.92	92.3%	56.2%	18.8%
4 7 Years	15	12	4.00	100.0%	73.3%	40.0%
8 15 Years	14	8	3.88	100.0%	42.9%	28.6%
16+ Years	29	18	3.94	100.0%	37.9%	10.3%

Objective 3: Perceived Training priorities

Objective 4: Training delivery preferences

Objective 5: Perceived workplace bottlenecks

Objective 6 : Innovation and innovation readiness

Conclusion and Recommendations

Key plots

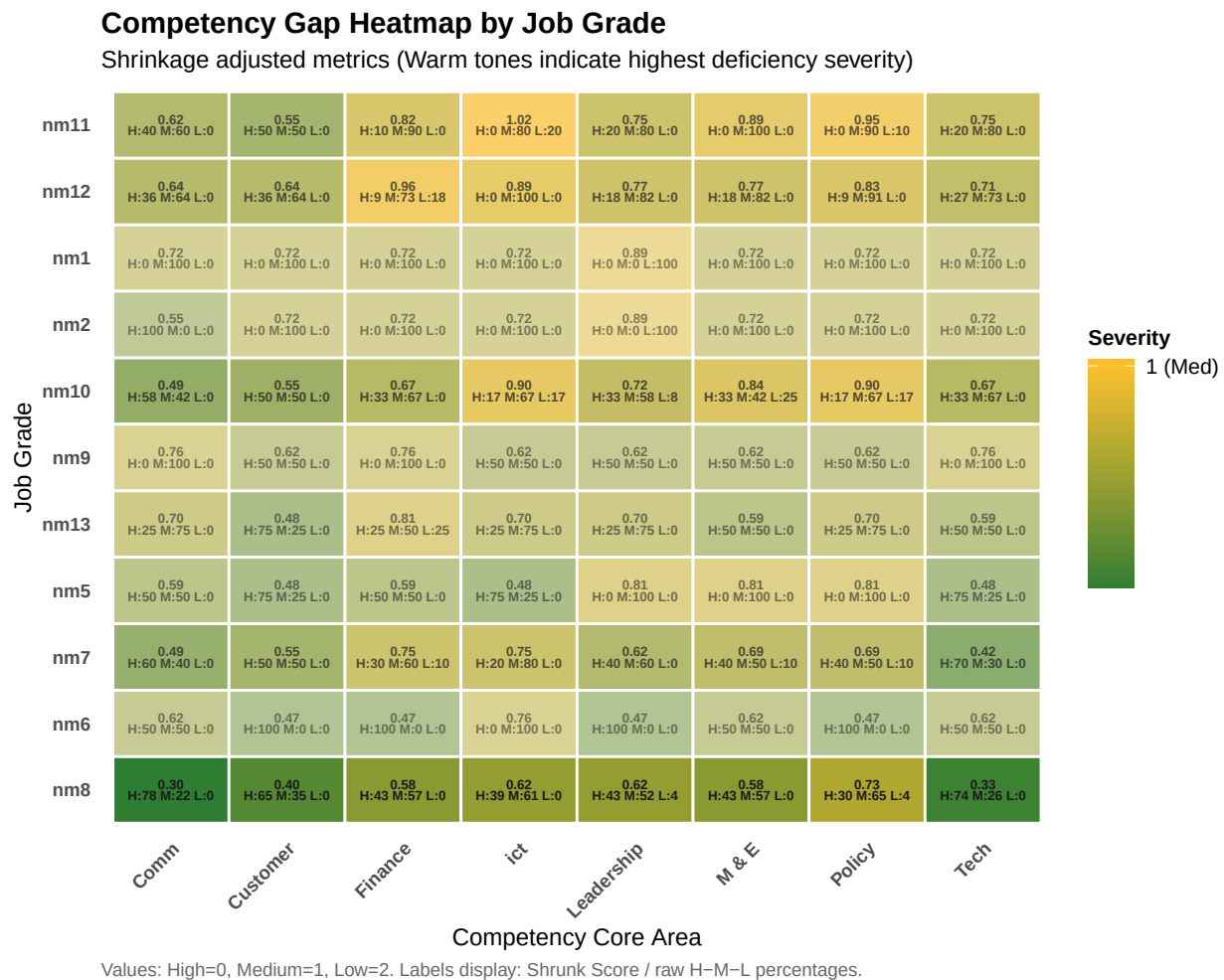


Figure 4: Competency Gap Heatmap by Job Grade Cadre.